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Analysis of urban morpho-emissivity in Kaunas

Geomatics and Spatial Modeling

Advanced Remote Sensing and Spatial Modeling

Teacher
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INTRODUCTION


Kaunas, the second largest city in Lithuania, is characterized by a contrasting urban structure: a dense historic center, modern residential neighborhoods, and industrial areas on the outskirts. In this context, urban forms, building materials, and vegetation cover directly influence heat exchanges and the formation of heat islands. The study is part of an approach to understanding urban thermal dynamics through the morphology and composition of surfaces, with a view to more sustainable planning.

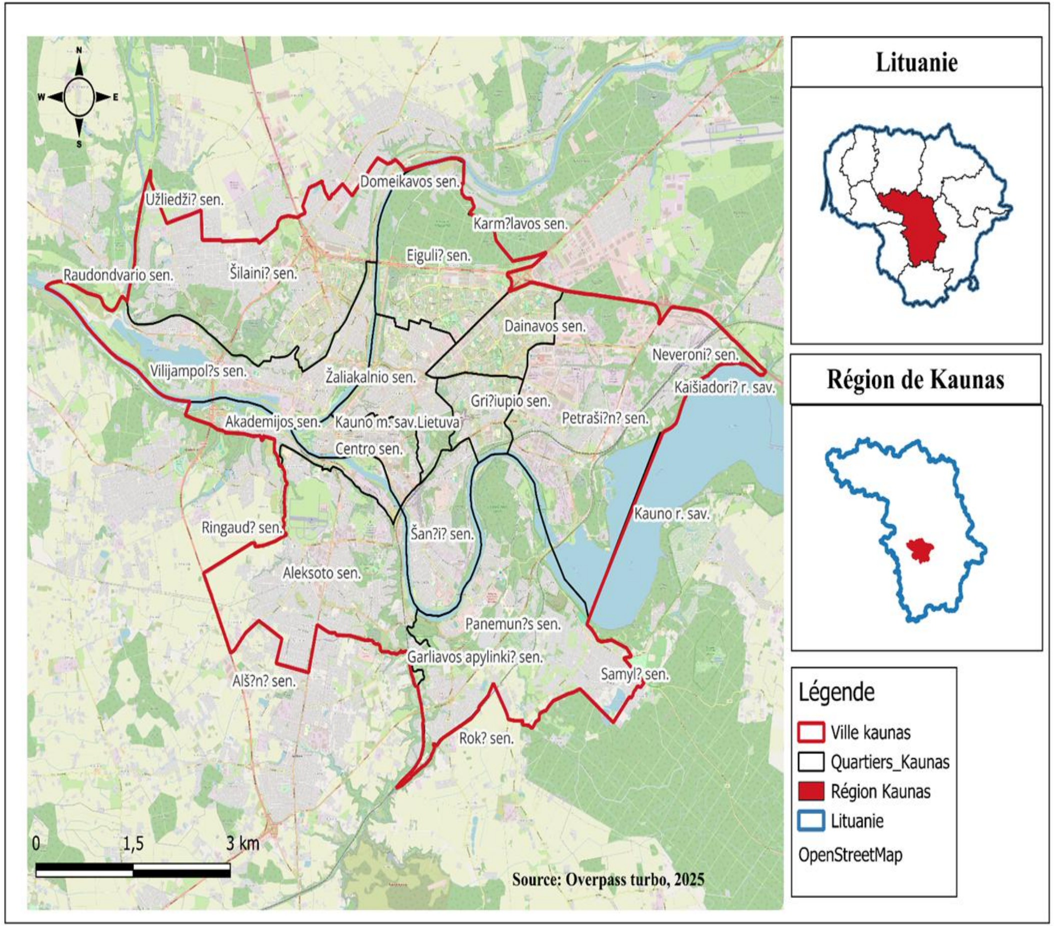
Scientific objectives

- Identify the relationships between land use, urban forms, materials, vegetation, and thermal emissivity.
- Set up a multi-scale spatial analysis model (neighborhood → urban block → building).
- Integrate complementary variables: age, function, and surrounding environment of the buildings.

Research questions

- How do urban form and the nature of materials influence surface temperature and emissivity?
- Which morphological parameters best explain thermal variations between neighborhoods?
- What is the contribution of vegetation to thermal regulation in Kaunas?

Study area presentation



- The map locates the city of Kaunas within Lithuania and its administrative region.
- The city boundary is shown in red, while neighborhoods (elderships) are delineated in black, highlighting the internal spatial structure used for multi-scale analysis.
- The Nemunas and Neris rivers strongly shape the urban organization and are key elements in environmental and climatic studies.

Data and sources

Satellite and remote sensing data

Source	Type	Resolution	Main use
Pléiades (dual scene)	Multispectral optical	0.5 m	Morphology, surface materials, fine vegetation
PRISMA	Hyperspectral (VNIR–SWIR)	30 m	Material identification, emissivity estimation
Landsat 8/9 (OLI/TIRS)	Optical + thermal	15–30 m	Land Surface Temperature (LST) calculation
OSM / BD Bâti	Vector data	Variable	Building footprints, height, and function

Additional data

- Administrative boundaries: elderships (districts) of Kaunas.
- Road network and land use (OSM).
- Contextual data (economic activities, green zones).

software

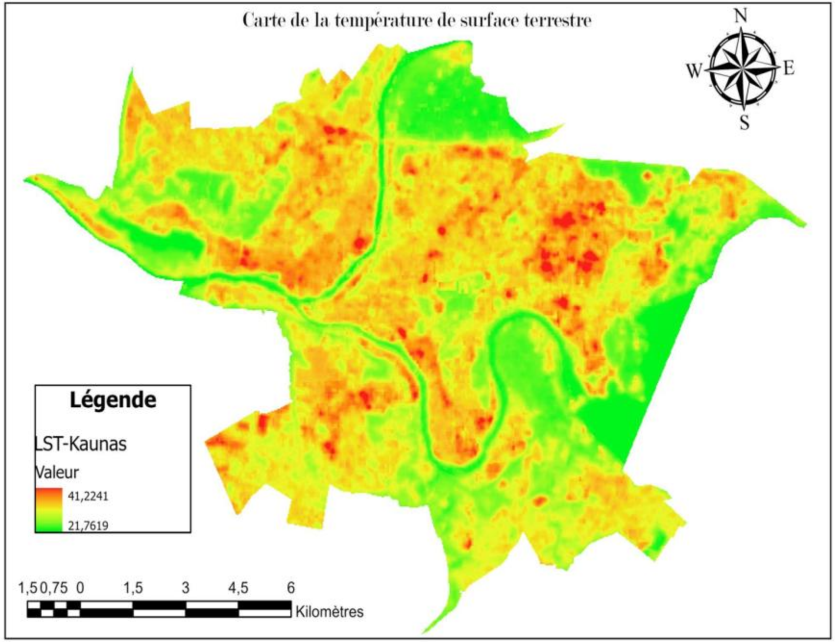
- QGIS / ArcGIS Pro: spatial processing, mapping.
- Python (rasterio, geopandas, momepy, mgwr): morphological calculations, regressions.
- Google Earth Engine (GEE): calculation of the indices NDVI, NDBI, MNDWI.
- FME Workbench: data cleaning and integration.

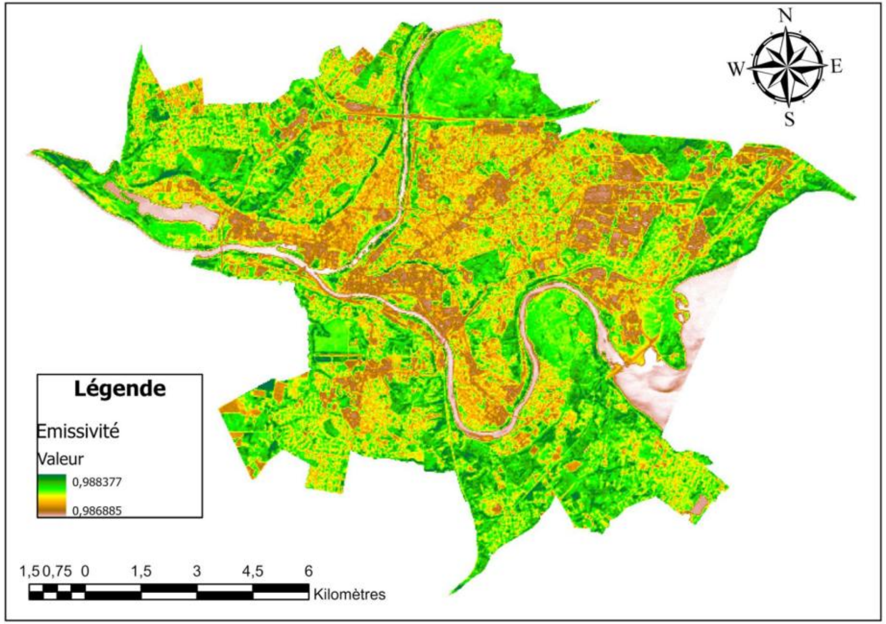
Geographical objects

Geographical Objects	Descriptions	Link to thermal behavior	Method of application
Urban Morphology (districts, blocks, building density)	Spatial organization of the built environment: form, compactness, density, building height.	Morphology affects heat retention, ventilation, and thermal contrasts across the city.	Compute indicators (BCR, FAR, compactness) and analyze emissivity and LST variations across spatial units.
Building Materials (concrete, asphalt, metal, tiles)	Composition of roofs and built surfaces.	Each material has a specific emissivity: metal → low, concrete/tiles → higher. Direct impact on surface temperature.	Perform supervised classification (PRISMA + Pleiades) and assign emissivity coefficients from spectral libraries.
Vegetation (trees, parks, green areas)	Vegetated surfaces scattered within the urban fabric.	Very high emissivity and strong cooling effect through evapotranspiration.	Compute NDVI, extract vegetated areas, and quantify green proportion per block or district.
Artificial Surfaces (roads, parking lots, asphalt)	Impervious infrastructures widely present in urban areas.	Medium emissivity but strong heat-storage capacity.	Detect through classification + NDBI. Integrate into emissivity mapping and LST analysis.

Earth surface temperature

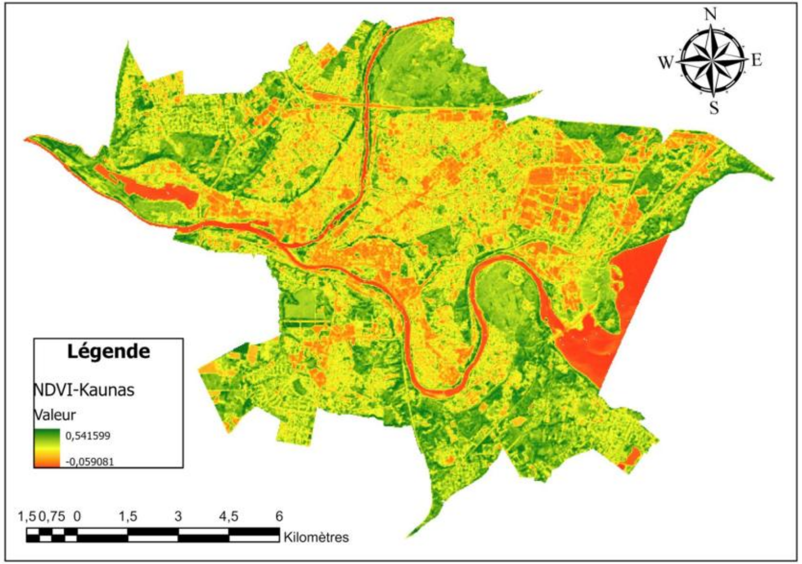
Emissivity of the Earth's surface



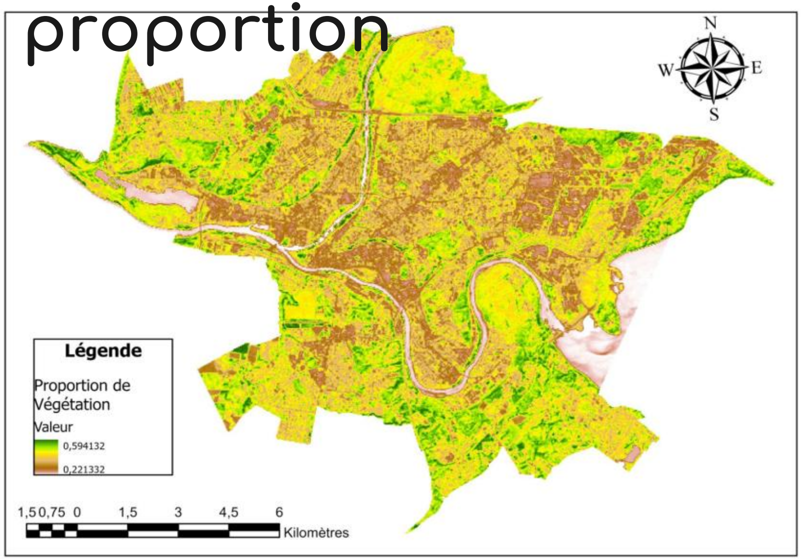


- Areas with high emissivity (vegetation, rivers, natural spaces) correspond to the lowest temperatures;
- These surfaces dissipate heat effectively, which explains their role as cool zones in the urban landscape;
- Areas with lower emissivity (dense downtown, industrial zones, mineral surfaces) show the highest temperatures;
- Urban materials (concrete, asphalt, metal) and the low presence of vegetation promote heat accumulation;
- This difference creates marked thermal contrasts at the heart of urban heat islands; the general relationship is clear: emissivity ↓ = temperature ↑, and emissivity ↑ = temperature ↓;
- This consistency confirms the direct link between materials, urban forms, and thermal dynamics in Kaunas.

NDVI-2024

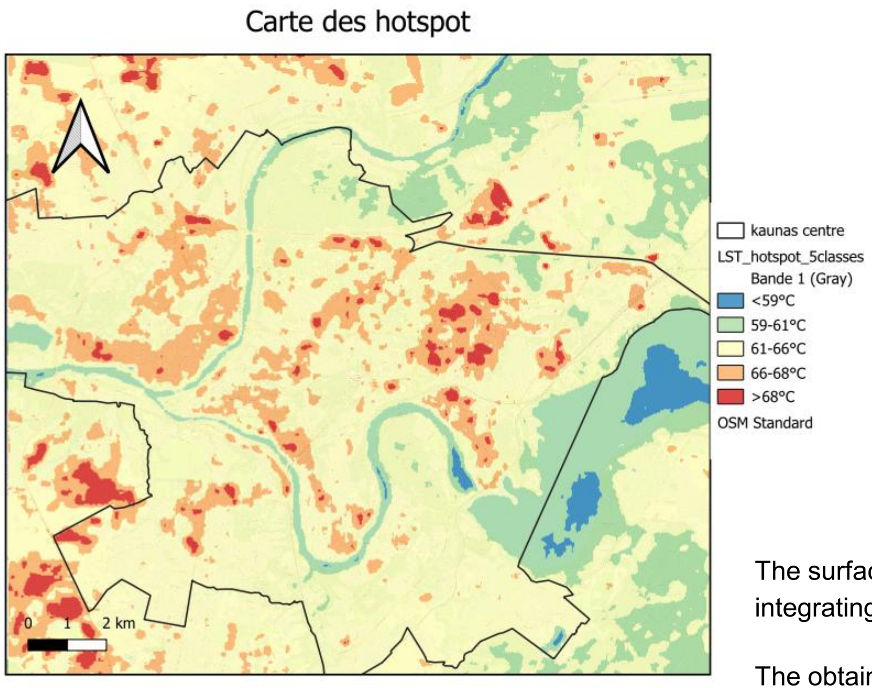


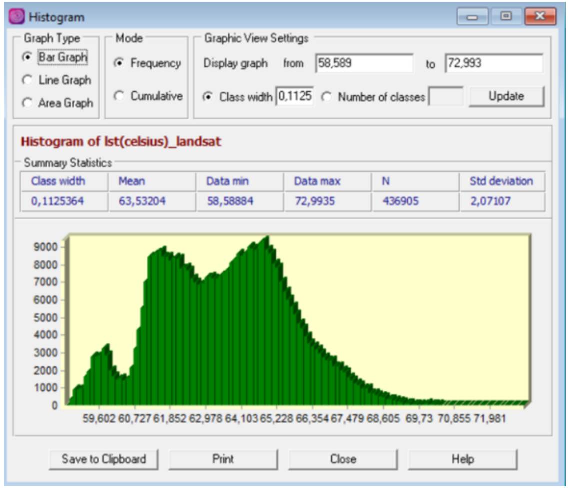
Vegetation proportion



- High NDVI values and vegetation proportions are concentrated in peripheral and forested areas, revealing a strong presence of plant cover;
- The downtown area and densely built-up sectors show low values on both maps, indicating limited vegetation availability;
- The river corridors of the Nemunas and Neris stand out as highly vegetated areas, playing a key role in natural cooling;
- Industrial zones consistently appear with low values, confirming the absence of vegetation and their potential contribution to heat zones;
- The correspondence between the two maps is clear: where NDVI is high, vegetation proportion is also high, which validates the consistency of the analyses.

Hotspot





The surface temperature (LST) was estimated from the thermal band of Landsat 8, integrating an emissivity correction derived from NDVI.

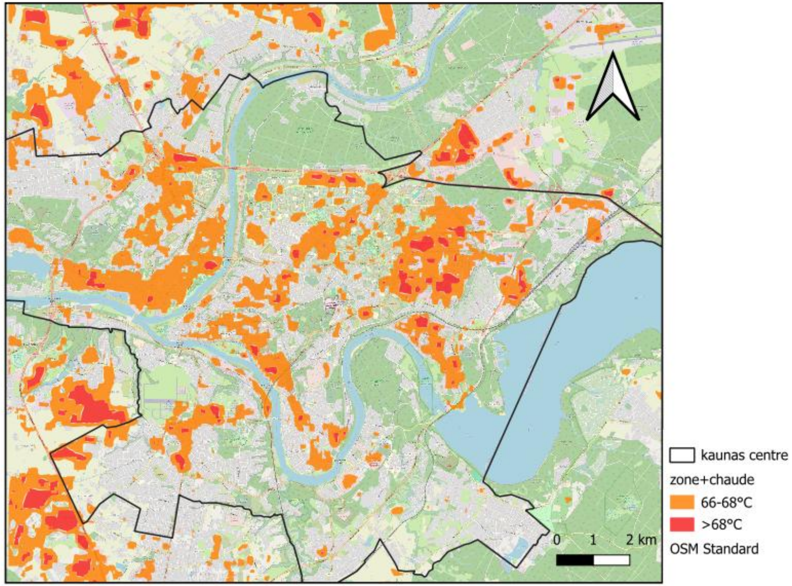
The obtained continuous map, centered on the urban center of Kaunas, was exported in QGIS to analyze the temperature distribution via the histogram.

From this distribution, thermal thresholds were defined and used in the raster calculator to reclassify the LST into five classes, ranging from the freshest areas (water and vegetation) to urban thermal hotspots, mainly associated with built-up and waterproofed surfaces.

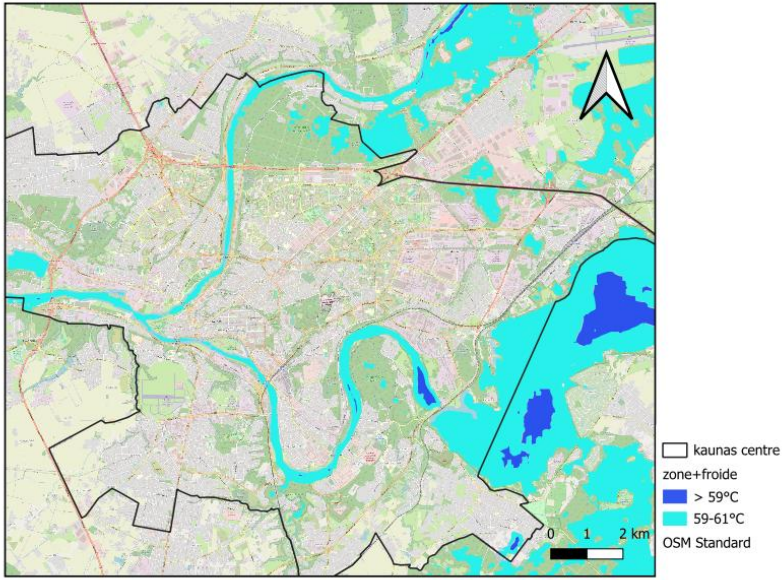
Hot and Cold areas

From the hotspot map, I extracted the two hottest classes and the two coldest classes in order to make separate maps showing the areas

Carte des zones les plus chaudes



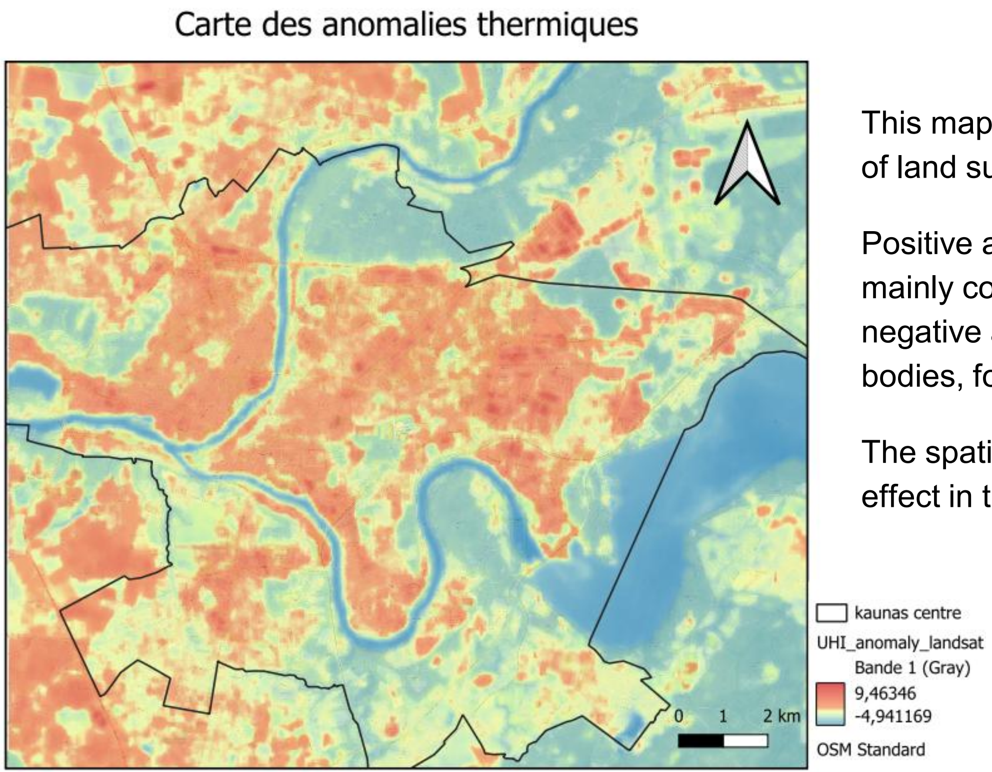
Carte des zones les plus froides



By superimposing, we can clearly see that the hottest areas correspond to urban and built-up areas.

While the cooler areas correspond mainly to aquatic surfaces and have some vegetated areas.

Thermal Anomaly Map

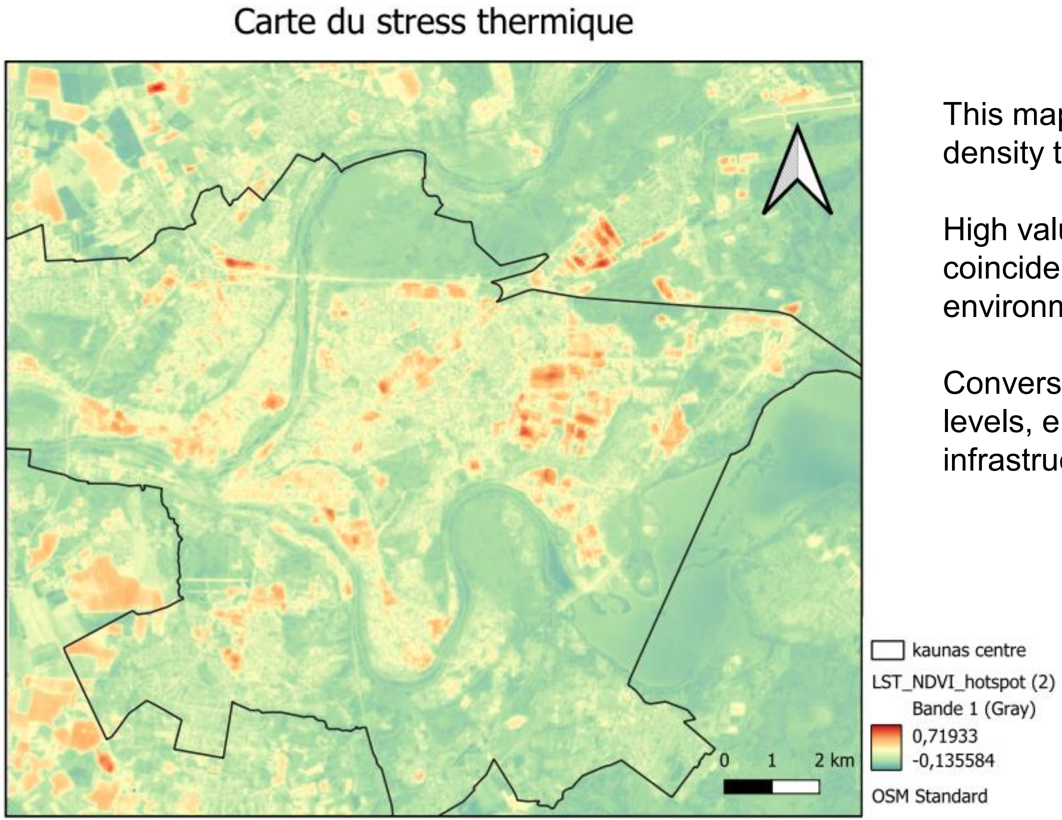


This map represents thermal anomalies by highlighting deviations of land surface temperature (LST) from the urban mean.

Positive anomalies indicate areas warmer than the city average, mainly concentrated in dense urban and industrial zones, while negative anomalies correspond to cooler areas such as water bodies, forests, and vegetated spaces.

The spatial pattern clearly reveals the Urban Heat Island (UHI) effect in the central parts of Kaunas.

Thermal Stress Map (LST × NDVI)

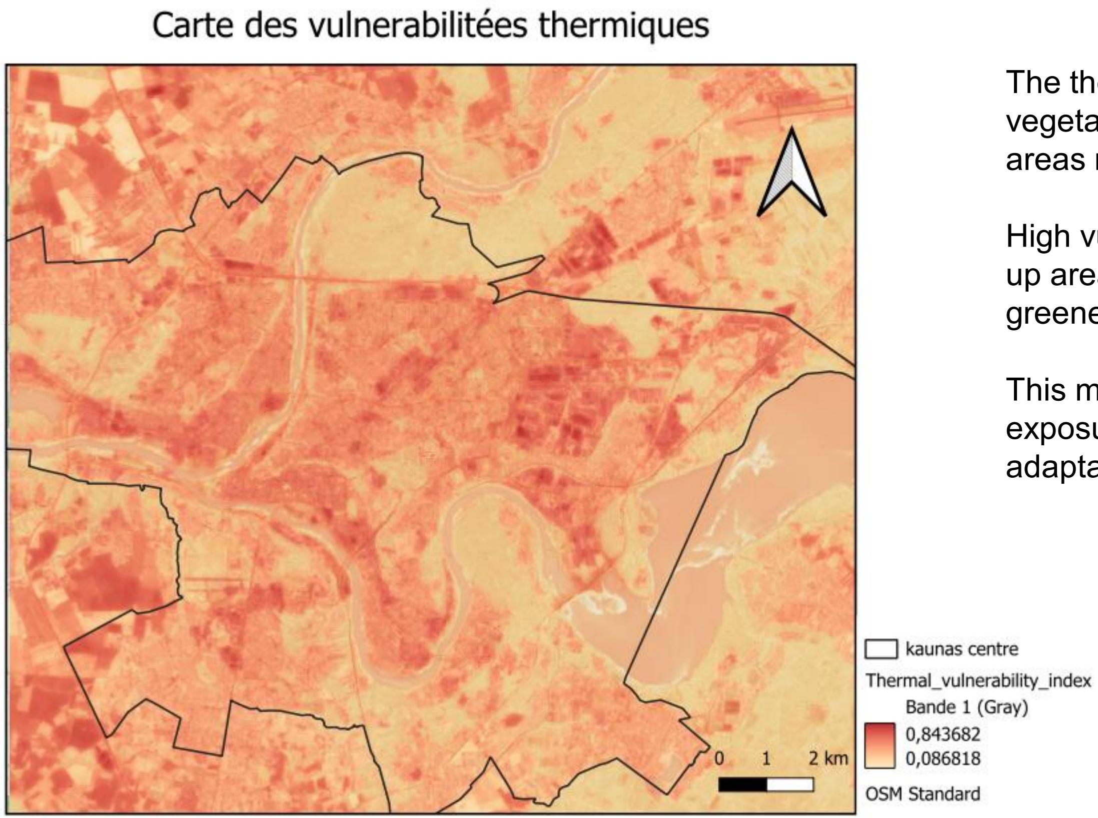


This map combines land surface temperature with vegetation density to identify areas experiencing thermal stress.

High values correspond to zones where high temperatures coincide with low vegetation cover, typical of built-up environments.

Conversely, vegetated and aquatic areas show lower stress levels, emphasizing the cooling role of green and blue infrastructures.

Thermal Vulnerability Map

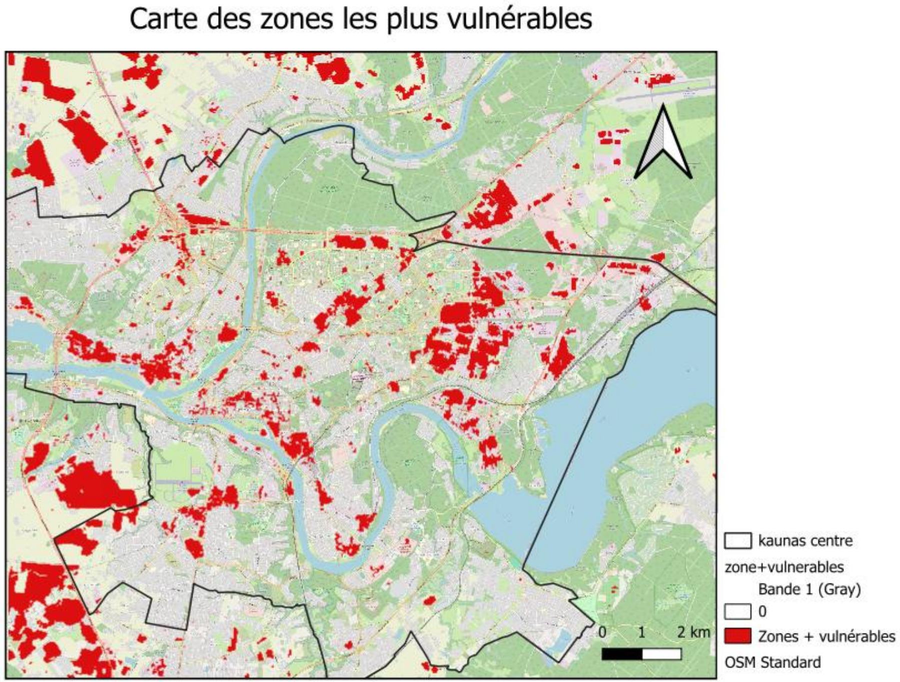


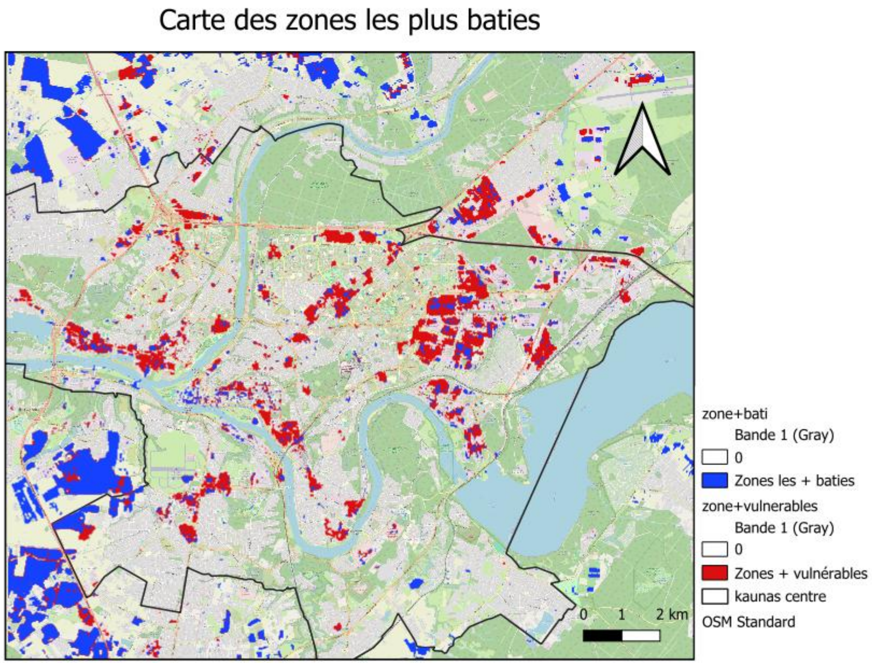
The thermal vulnerability map integrates temperature, vegetation deficit, and urban characteristics to highlight areas most exposed to heat-related risks.

High vulnerability zones are mainly located in densely built-up areas with limited vegetation, whereas peripheral and greener zones appear less vulnerable.

This map provides a synthetic view of spatial inequalities in exposure to urban heat and can support urban climate adaptation strategies.

Vulnerability x buildings

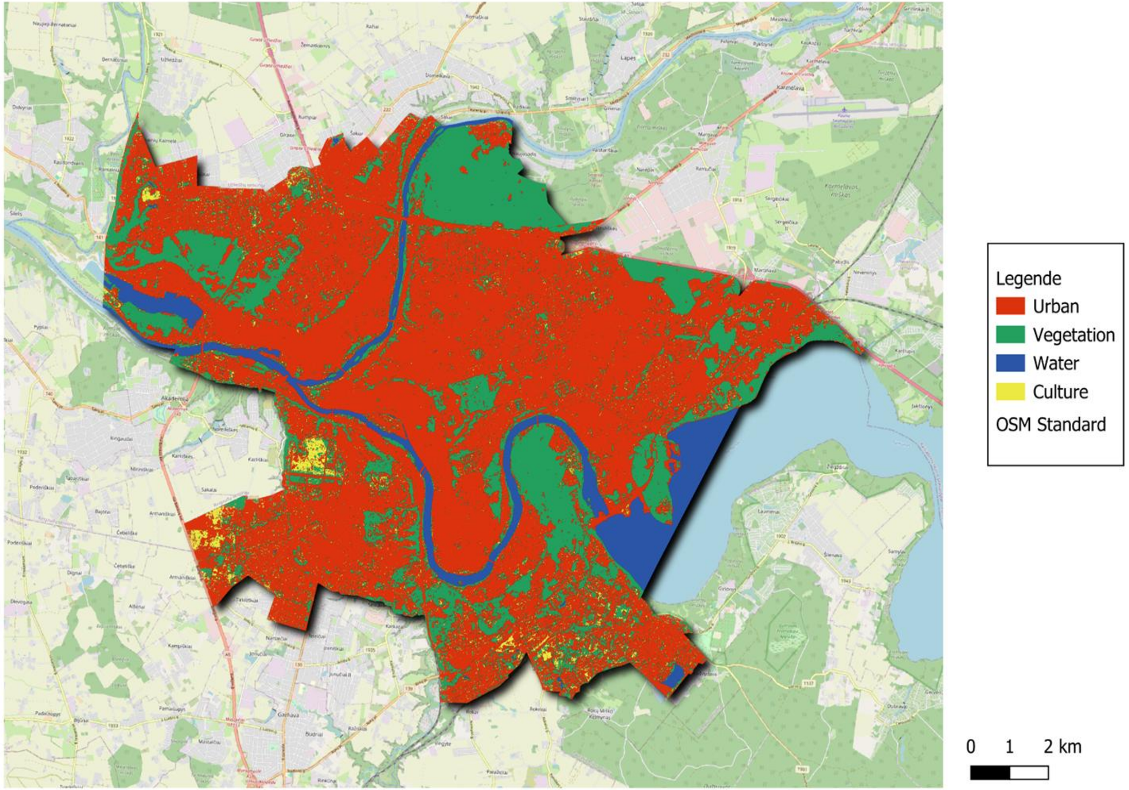




By extracting the most thermally vulnerable areas and overlaying them with the most built-up zones (derived from the NDBI), a clear spatial correspondence emerges.

The highest vulnerability levels are mainly concentrated in densely urbanized areas, which is consistent with the physical properties of built surfaces. Buildings and impervious materials store and re-emit heat, limit evapotranspiration, and reduce surface cooling, thereby reinforcing

Landcover Kaunas 2024



The map shows urbanisation (red) dominating most of the territory, while vegetation (green) is concentrated in small scattered patches.

The hydrographic network (blue) follows the main river valleys. Cultivated areas (yellow) are located on the urban peripheries. This mosaic landscape reveals remarkable urban intensification at the expense of natural spaces.

This land cover map of Kaunas (2024) provides the essential context for thermo-emissivity analysis, as emissivity properties vary greatly depending on land cover classes (urban, vegetation, crops, water). The dominant urbanisation (red) has high emissivity , promoting heat retention and an urban heat island effect (UHI), unlike vegetation (green) and water (blue), which have lower and variable emissivities, which moderate surface temperature (LST).